Teacher answer key: Primary 4, Semester 1 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 4 content in the Mathematics Syllabus, Primary (2021 edition, pages 37 to 40). The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 1 and 2)

Strand	Pages
Number and Algebra - Whole Numbers	S1-01, S1-02, S1-03, S1-04, S1-05, S1-06, S1-07, S1-08, S1-09, S1-10, S1-11, S1-12, S1-13
Number and Algebra - Fractions	S1-14, S1-15, S1-16, S1-17, S1-20
Measurement and Geometry - Geometry	S1-18, S1-19

The page code in the footer, for example S1-07 (P4-07), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S1-01 (P4-01) Place Value 1 star

Topic: Place value: ten thousands, thousands, hundreds, tens and ones

Syllabus outcomes: Numbers up to 100 000 1.1, Numbers up to 100 000 1.2. Notes on the tag: 1.1 (number notation, representations and place values), 1.2 (reading and writing numbers)
Puzzle type: Code-breaker (place value). How the pupil checks: Code word (6 letters).
Answers: 1) chart: 3 ten thousands, 13 thousands, 6 hundreds, 0 tens, 2 ones = 43 602; 2) 3000; 3) 80; 4) 31 510; 5) 60 000; 6) 30 466
Check result: Code word: HUNGRY
Teacher note: A place-value chart whose thousands column holds more than 9 (regrouping), the value of a digit, the number of hundreds in a number, 10 000 more, the value of a digit in the ten-thousands place and the sum of four parts given in a mixed-up order. Decoy letters stand for slips such as giving the digit instead of its value, a wrong place, or forgetting to regroup. A box left blank gives no letter and the check cannot see it.

S1-02 (P4-02) Words and Numerals 1 star

Topic: Reading and writing numbers up to 100 000 in numerals and in words

Syllabus outcomes: Numbers up to 100 000 1.2. Notes on the tag: 1.2 (reading and writing numbers in numerals and in words)
Puzzle type: Matching (number words to numeral cards). How the pupil checks: Word from the letters of the matching cards (6 letters).
Answers: 1) fifty-two thousand, six hundred and sixty = 52 660 (card B); 2) forty-two thousand and seventy-nine = 42 079 (card A); 3) forty-one thousand and sixty-three = 41 063 (card C); 4) twelve thousand, nine hundred and seven = 12 907 (card K); 5) sixty-seven thousand, eight hundred and fifty = 67 850 (card U); 6) twenty thousand and eighty = 20 080 (card P)
Check result: Word: BACKUP
Teacher note: Six numbers written in words (thousands, hundreds, tens and ones, with a zero in the middle of some). Ten numeral cards are printed: six match, four are decoys made by swapping two digits or dropping a zero (the usual slips when a number in words is written as a numeral). The letters of the matching cards, read down the six rows, spell the word. A decoy card put in any row gives a letter that does not spell a word, and a swap of two rows does not either (all tested against a very large word list). The pupil writes a letter, not a number, so writing a number word as numerals by hand is not asked. A box left blank gives no letter and the check cannot see it.

S1-03 (P4-03) Number Patterns 2 stars

Topic: Number patterns: find the missing number

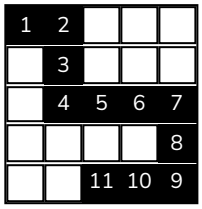
Syllabus outcomes: Numbers up to 100 000 1.4. Notes on the tag: 1.4 (patterns in number sequences)
Puzzle type: Code-breaker (number patterns). How the pupil checks: Code word (6 letters).
Answers: 1) 25 300; 2) 43 500; 3) 4800; 4) 12 500; 5) 9800; 6) 10 000
Check result: Code word: CREAMY
Teacher note: Six patterns with numbers up to 100 000: a step up or down with the last number missing, a step down with a number missing in the middle, a missing first number, a pattern that goes up by one amount and down by another in turn (the next number is asked), the 8th, 9th or 12th number found from the first number and the step, and a doubling pattern. The decoy letters stand for the usual slips: half the step, one step too many or too few, or forgetting the take-away step. A box left blank gives no letter and the check cannot see it.

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S1-04 (P4-04) Round It 1 star

Topic: Rounding numbers to the nearest 10, 100 or 1000; the sign for about

Syllabus outcomes: Numbers up to 100 000 1.5, Numbers up to 100 000 1.6. Notes on the tag: 1.5, 1.6 (rounding numbers to the nearest 10, 100 or 1000; use of the sign for about)
Puzzle type: Maze / path (rounding, true statements). How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.

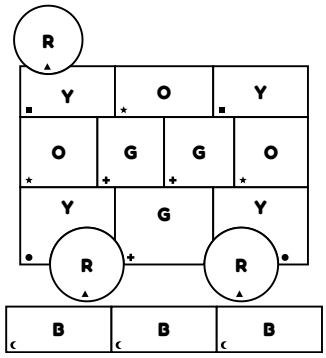


Answers: Route (row, column from top-left): (1,1) 44 220 = 44 200 to 100 -> (1,2) 80 443 = 80 000 to 1000 -> (2,2) 86 324 = 86 320 to 10 -> (3,2) 49 533 = 50 000 to 1000 -> (3,3) 46 476 = 46 500 to 100 -> (3,4) 37 816 = 37 820 to 10 -> (3,5) 29 783 = 29 780 to 10 -> (4,5) 34 038 = 34 040 to 10 -> (5,5) 29 500 = 30 000 to 1000 -> (5,4) 73 523 = 74 000 to 1000 -> (5,3) 40 029 = 40 000 to 100
Check result: Route reaches FINISH through 11 boxes.
Teacher note: A 5 by 5 grid of boxes, each with a number, the sign for about, a rounded number and the place it is rounded to (10, 100 or 1000). On the route every statement is true. Boxes off the route are false: they give the number rounded down when it should round up, up when it should round down, or one step too far. 3 extra boxes with true statements are dead ends. Counting the boxes shows a line that takes a shortcut. Pupils round half way numbers up. The pupil reads the sign for about as is about.

S1-05 (P4-05) Colour the Bus 2 stars

Topic: Numbers up to 100 000: place value, order, rounding

Syllabus outcomes: Numbers up to 100 000 1.1, Numbers up to 100 000 1.2, Numbers up to 100 000 1.3, Numbers up to 100 000 1.4, Numbers up to 100 000 1.5. Notes on the tag: 1.1, 1.2, 1.3, 1.4, 1.5 (place value, numbers in words, comparing, patterns and rounding; numbers up to 100 000)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.



Answers: Colour key: 400 = yellow, 3000 = red, 4500 = blue, 7000 = green, 20 000 = orange
Check result: Tally: yellow 4, red 3, blue 3, green 3, orange 3; parts with the same small mark match.

Teacher note: Every part asks a question about numbers up to 100 000: the value of a digit, a number in words, 10, 100, 1000 or 10 000 more or less, the greatest or smallest of three close numbers, a number pattern or rounding to the nearest 10, 100 or 1000. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

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S1-06 (P4-06) Factor or Not? 1 star

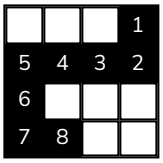
Topic: Factors and multiples: is it a factor? is it a multiple?

Syllabus outcomes: Factors and Multiples 2.1, Factors and Multiples 2.2, Factors and Multiples 2.4. Notes on the tag: 2.1 (factors, multiples and their relationship), 2.2 (a 1-digit number as a factor of a number within 100), 2.4 (a multiple of a 1-digit number)
Puzzle type: Choice table (yes or no). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) Yes (Is 8 a factor of 32?); 2) No (Is 8421 a multiple of 8?); 3) Yes (Is 4 a factor of 32?); 4) Yes (Is 2872 a multiple of 4?); 5) Yes (64 is a multiple of 8. Is 8 a factor of 64?); 6) No (84 is a multiple of 7. Is 84 a factor of 7?)
Check result: Word: TEACUP
Teacher note: Six yes or no questions: two ask whether a 1-digit number is a factor of a number within 100, two ask whether a 4-digit number is a multiple of a 1-digit number (the pupil divides), and two start from a true statement and ask the same fact the other way round or reversed (the relationship between factor and multiple). Each row has a Yes and a No, each with a letter. The letters of the right answers, read down the rows, spell the word; every other way of choosing was tested against a very large word list and none spells a word. A box left blank gives no letter and the check cannot see it.

S1-07 (P4-07) Multiples Maze 1 star

Topic: Multiples of a 1-digit number

Syllabus outcomes: Factors and Multiples 2.1, Factors and Multiples 2.4. Notes on the tag: 2.4 (determining if a number is a multiple of a given 1-digit number), 2.1 (factors, multiples and their relationship)
Puzzle type: Maze / path (multiples). How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
Answers: Route (row, column from top-left): (1,4) 6258 = 7 x 894 -> (2,4) 7210 = 7 x 1030 -> (2,3) 9933 = 7 x 1419 -> (2,2) 7077 = 7 x 1011 -> (2,1) 7315 = 7 x 1045 -> (3,1) 4025 = 7 x 575 -> (4,1) 6685 = 7 x 955 -> (4,2) 9520 = 7 x 1360
Check result: Route reaches FINISH through 8 boxes.



Teacher note: A 4 by 4 grid of 4-digit numbers. On the route every number is a multiple of 7 (the pupil divides). The boxes off the route are numbers near a multiple of 7 (one to three more or fewer) or a multiple with two digits swapped, so a slip in dividing sends the pupil there. 2 extra boxes that are multiples of 7 are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-08 (P4-08) Common Factors 2 stars

Topic: Common factors of two numbers

Syllabus outcomes: Factors and Multiples 2.1, Factors and Multiples 2.3 (problem solving). Notes on the tag: 2.3 (finding the common factors of two given numbers), 2.1 (factors, multiples and their relationship)
Puzzle type: Code-breaker (common factors). How the pupil checks: Code word (6 letters).
Answers: 1) 11; 2) 4; 3) 12; 4) 15; 5) 8; 6) 7
Check result: Code word: FROZEN
Teacher note: Six questions about the common factors of two numbers within 100: the greatest common factor, how many common factors, the sum of all the common factors (the pupil must find every one), the longest equal pieces of two ribbons, the second greatest common factor and the one common factor between two limits. Decoy letters stand for slips such as giving the number of factors of one number only, a common multiple, or missing a common factor. A box left blank gives no letter and the check cannot see it.

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S1-09 (P4-09) Number Riddles 2 stars

Topic: Factors and multiples: who am I?

Syllabus outcomes: Factors and Multiples 2.1, Factors and Multiples 2.3, Factors and Multiples 2.4, Factors and Multiples 2.5 (problem solving). Notes on the tag: 2.1 (factors, multiples and their relationship), 2.3 (common factors), 2.4 (multiples), 2.5 (common multiples of two 1-digit numbers)
Puzzle type: Number riddles. How the pupil checks: Code word (6 letters).
Answers: 1) 54; 2) 80; 3) 98; 4) 43; 5) 48; 6) 186
Check result: Code word: PEANUT
Teacher note: Six riddles with three clues each, drawn from: a multiple of a 1-digit number, a factor of a number within 100, a common multiple of two 1-digit numbers, a common factor of two numbers within 100, greater than, less than, odd or even and the ones digit. Each riddle was solved by trying every whole number from 1 to 999: exactly one number fits all three clues. The decoy letters stand for numbers that fit two of the three clues. A box left blank gives no letter and the check cannot see it.

S1-10 (P4-10) Long Multiplication 1 star

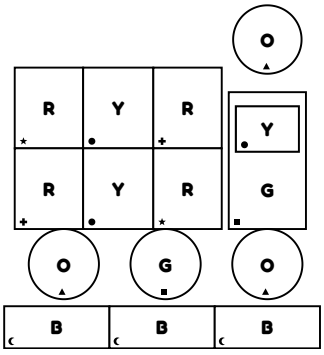
Topic: Multiplying: 4 digits by 1 digit, 3 digits by 2 digits

Syllabus outcomes: Four Operations 3.1. Notes on the tag: 3.1 (multiplication algorithm: up to 4 digits by 1 digit; up to 3 digits by 2 digits)
Puzzle type: Code-breaker (multiplication algorithm). How the pupil checks: Code word (6 letters).
Answers: 1) 2905 x 9 = 26 145; 2) 4773 x 3 = 14 319; 3) 3039 x 6 = 18 234; 4) 423 x 49 = 20 727; 5) 161 x 73 = 11 753; 6) 485 x 28 = 13 580
Check result: Code word: FRUITY
Teacher note: Three multiplications of 4 digits by 1 digit and three of 3 digits by 2 digits, each needing renaming (carrying). The decoy letters stand for the usual slips: forgetting to carry, and adding the two partial products of a 3-digit by 2-digit sum without shifting the second one. A box left blank gives no letter and the check cannot see it.

S1-11 (P4-11) Colour the Lorry 2 stars

Topic: Multiplication and division with larger numbers

Syllabus outcomes: Four Operations 3.1, Four Operations 3.2. Notes on the tag: 3.1, 3.2 (multiplication algorithm up to 4 digits by 1 digit and 3 digits by 2 digits; division algorithm up to 4 digits by 1 digit)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.
Answers: Colour key: 1260 = yellow, 1680 = red, 2520 = blue, 3360 = green, 5040 = orange
Check result: Tally: yellow 3, red 4, blue 3, green 2, orange 3; parts with the same small mark match.



Teacher note: Every part asks a multiplication or division with larger numbers: 4 digits by 1 digit, 3 digits by 2 digits, a missing factor, a division of a 4-digit number by a 1-digit number, a short story or a multiplication followed by a subtraction. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

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S1-12 (P4-12) Division 2 stars

Topic: Division: 4 digits divided by 1 digit

Syllabus outcomes: Four Operations 3.2 (problem solving). Notes on the tag: 3.2 (division algorithm, up to 4 digits by 1 digit)
Puzzle type: Code-breaker (division algorithm). How the pupil checks: Code word (6 letters).
Answers: 1) 2086; 2) 685; 3) 1; 4) 314; 5) 354; 6) 824
Check result: Code word: SHRIMP
Teacher note: Six divisions of a 4-digit number by a 1-digit number: three plain (one with a zero in the answer, one exact and one that asks only for the remainder) and three short stories (share equally, how many days, how many full packs with a remainder). The decoy letters stand for the usual slips: leaving a zero out of the answer, giving the quotient where the remainder was asked, or adding 1 for the left-over stickers when the question asks for full packs. A box left blank gives no letter and the check cannot see it.

S1-13 (P4-13) Number Ring 2 stars

Topic: Multiplying and dividing: round the ring and back to the start

Syllabus outcomes: Four Operations 3.1, Four Operations 3.2 (problem solving). Notes on the tag: 3.1 (multiplication algorithm), 3.2 (division algorithm)
Puzzle type: Chain loop (operations). How the pupil checks: Loop: following every arrow brings the pupil back to the start number.
Answers: Circles after the start, clockwise: 1432, 716, 4296, 1074, 358. Arrows: ×8, ÷2, ×6, ÷4, ÷3, ÷2
Check result: Going round the ring ends on the start number.
Teacher note: A ring of six numbers. The pupil starts at 179 and follows five arrows, each a multiplication or a division by a 1-digit number, writing every new number. The sixth arrow takes the last number back to 179. Every number the pupil writes is used by the next arrow, so one slip makes the ring fail to close. All the numbers stay below one hundred thousand. The multiplications use the algorithm up to 4 digits by 1 digit and the divisions are exact.

S1-14 (P4-14) Mixed or Improper 1 star

Topic: Mixed numbers and improper fractions

Syllabus outcomes: Mixed Numbers and Improper Fractions 1.1. Notes on the tag: 1.1 (mixed numbers, improper fractions and their relationship)
Puzzle type: Code-breaker (mixed numbers and improper fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 9/5; 2) whole number 1 (13/8); 3) 19/4 = 4 3/4, whole number 4; 4) 5 4/8 = 44/8; 5) 48/7 = 6 6/7, top number 6; 6) 5 = 20/4
Check result: Code word: FIGURE
Teacher note: Two pictures of fraction bars (the pupil counts the shaded parts, then writes the improper fraction or the whole number of the mixed number), a conversion of an improper fraction to a mixed number (the whole number is missing), of a mixed number to an improper fraction (the top number is missing), the fraction part of a mixed number, and a whole number written as a fraction. The decoy letters stand for the usual slips: adding the whole number to the top number, or giving the remainder where the whole number was asked. A box left blank gives no letter and the check cannot see it.

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S1-15 (P4-15) Fraction Number Line 1 star

Topic: Mixed numbers and improper fractions on a number line

Syllabus outcomes: Mixed Numbers and Improper Fractions 1.1. Notes on the tag: 1.1 (mixed numbers, improper fractions and their relationship)
Puzzle type: Choice table (number lines). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 3/2 = 1 1/2; 2) 11/4 = 2 3/4; 3) 5/2 = 2 1/2; 4) 7/8 = 0 7/8; 5) 15/4 = 3 3/4; 6) 11/4 = 2 3/4
Check result: Word: FABRIC
Teacher note: Six number lines, each three units long, from 0 or from 1, marked in equal parts (the printed lines use halves, quarters and eighths) with an arrow. The pupil reads the arrow and picks the matching number from three, written as a mixed number or an improper fraction. The two wrong choices are the next or the previous mark, a whole number that is one too many or too few, the fraction part without the whole number, or the part counted from the other end. The letters of the right choices, read down the rows, spell the word; every other way of choosing was tested against a very large word list and none spells a word. A box left blank gives no letter and the check cannot see it.

S1-16 (P4-16) Fraction of a Set 1 star

Topic: Fraction as part of a set

Syllabus outcomes: Fraction of a Set 2.1 (problem solving). Notes on the tag: 2.1 (fraction as part of a set)
Puzzle type: Code-breaker (fraction of a set). How the pupil checks: Code word (6 letters).
Answers: 1) 2 of 4 rings of 3 = 6; 2) 2 of 3 rings of 4 = 8; 3) 15; 4) 18; 5) 12; 6) 28
Check result: Code word: WALRUS
Teacher note: Two pictures of a set in rings of equal size (the pupil finds the fraction by counting rings), a fraction of a number, a fraction of a set when the rest is asked, the whole set when a part is given, and a story where the rest is asked. The decoy letters stand for the usual slips: giving the rest instead of the part, one equal part instead of several, or the whole set. A box left blank gives no letter and the check cannot see it.

S1-17 (P4-17) Fraction Sums 2 stars

Topic: Adding and subtracting fractions with different denominators

Syllabus outcomes: Fractions: Addition and Subtraction 3.1, Mixed Numbers and Improper Fractions 1.1. Notes on the tag: 3.1 (adding and subtracting fractions with denominators not exceeding 12 and not more than two different denominators), 1.1 (improper fractions)
Puzzle type: Code-breaker (fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 7/10; 2) 1/5; 3) 13/10; 4) 1/10; 5) 1/2; 6) 8/9
Check result: Code word: BANDIT
Teacher note: Six sums and differences of two proper fractions: four with related denominators (one is a multiple of the other), two with denominators that are not related (two different denominators in all). One sum is above 1 and stays an improper fraction, and at least two answers need simplifying. The pupil writes the answer in simplest form in two boxes and finds it in the code table of fractions. The decoy letters stand for the usual slips: adding the tops and the bottoms, or leaving the answer unsimplified. A box left blank gives no letter and the check cannot see it.

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S1-18 (P4-18) Name the Angle 1 star

Topic: Naming angles with three letters, as in angle ABC

Syllabus outcomes: Angles 1.1. Notes on the tag: 1.1 (using notation such as ∠ABC and ∠a to name angles)
Puzzle type: Choice cards (angle names). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) angle SXL; 2) angle SZX; 3) angle XRQ; 4) angle QLR; 5) angle XQL; 6) angle SNP
Check result: Word: CURFEW
Teacher note: Six drawings with three lines from one point. An arc marks one angle; the three names offered are the three angles the lines make, each written with the point in the middle (for example angle PQR). Two drawings label the marked angle with the small letter a, as in angle a. The pupil must pick the name that matches the two lines the arc joins. The letters of the right choices, read down the cards, spell the word; every other way of choosing was tested against a very large word list and none spells a word. Drawing an angle of a given size is not asked on this page. A box left blank gives no letter and the check cannot see it.

S1-19 (P4-19) Protractor Readings 2 stars

Topic: Measuring angles in degrees

Syllabus outcomes: Angles 1.2. Notes on the tag: 1.2 (measuring angles in degrees)
Puzzle type: Code-breaker (protractor). How the pupil checks: Code word (6 letters).
Answers: 1) 130 degrees (base arm to the left); 2) 70 degrees (base arm to the left); 3) 110 degrees (base arm to the right); 4) 90 degrees (base arm to the right); 5) 50 degrees (base arm to the right); 6) 30 degrees (base arm to the left)
Check result: Code word: FAMILY
Teacher note: Six angles drawn on a protractor with a scale from each side, one arm along the base line, pointing to the right on three drawings and to the left on three. The pupil starts counting from 0 on the arm along the base, so a drawing with the base arm on the left must be read on the other scale. The decoy letters stand for readings 10 degrees out and for 180 minus the angle where that is free (the wrong scale). Every angle is 10 degrees past a multiple of 20 (30, 50, 70 and so on), so an arm lies 10 degrees from the nearest printed number and never over one. The page measures reading a protractor; drawing an angle with a protractor cannot be checked on a printed page. A box left blank gives no letter and the check cannot see it.

S1-20 (P4-20) Fraction Stories 2 stars

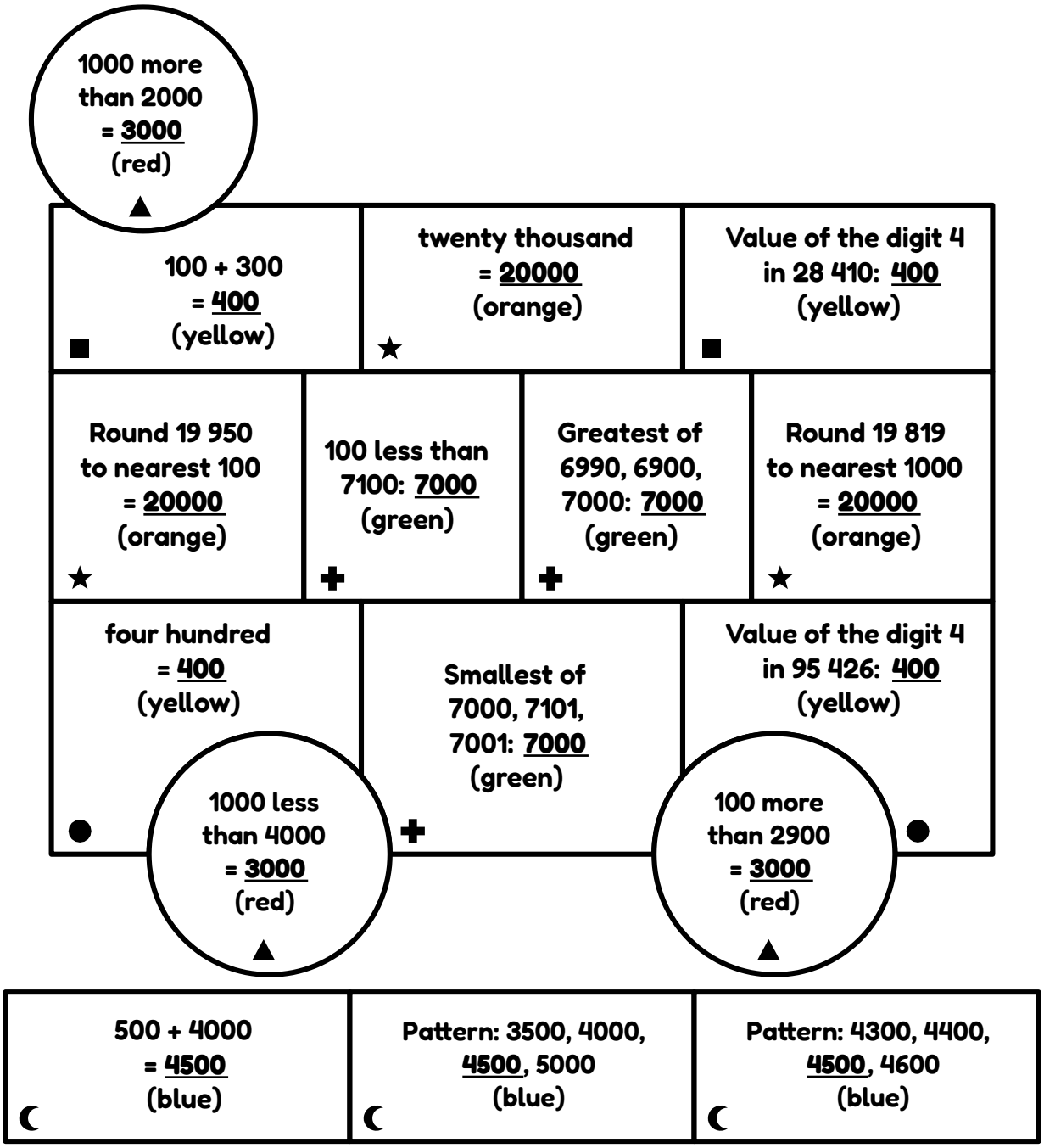
Topic: Fractions of a set in stories with two steps

Syllabus outcomes: Fraction of a Set 2.1, Fractions: Addition and Subtraction 3.1 (problem solving). Notes on the tag: 2.1 (fraction as part of a set), 3.1 (adding and subtracting fractions)
Puzzle type: Code-breaker (fraction stories). How the pupil checks: Code word (6 letters).
Answers: 1) 18; 2) 10; 3) 150; 4) 20; 5) 32; 6) 12
Check result: Code word: POLICE
Teacher note: Six stories with two steps: the parts given away from a whole set and the number left, what is in a tank that is part empty, two parts of a book read and the pages left, the whole number from a fraction of it, the whole amount from two spent fractions of it and the money left, and a fraction of the rest. The decoy letters stand for the usual slips: stopping after the first step, giving the part instead of the rest, or taking the second fraction from the whole instead of from the rest. A box left blank gives no letter and the check cannot see it.

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Answers: S1-05 (P4-05) Colour the Bus

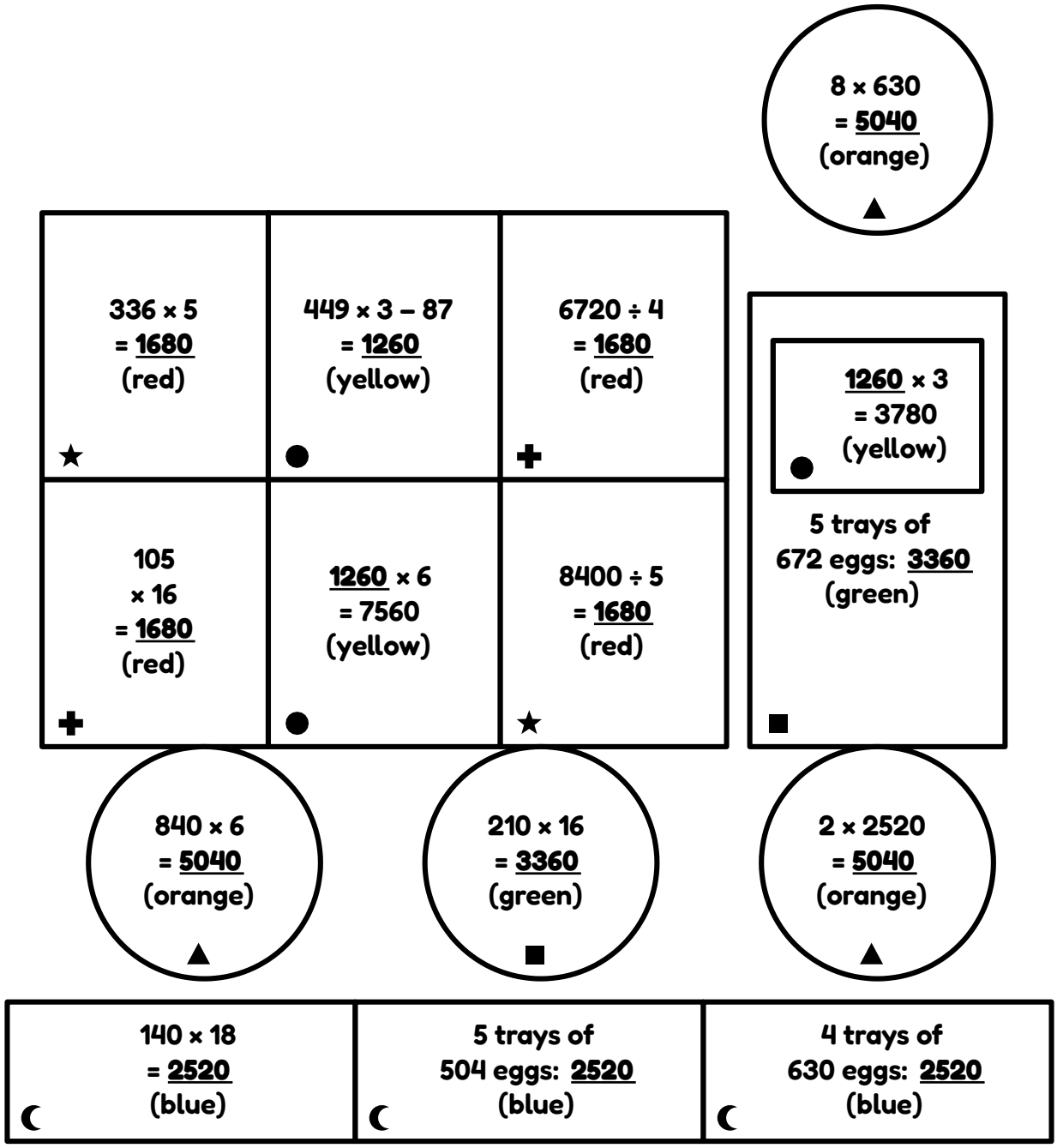
Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 400 = yellow, 3000 = red, 4500 = blue, 7000 = green, 20 000 = orange.



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Answers: S1-11 (P4-11) Colour the Lorry

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 1260 = yellow, 1680 = red, 2520 = blue, 3360 = green, 5040 = orange.



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